COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

B. TECH DEGREE II SEMESTER SECOND INTERNAL EXAMINATIONS, JULY 2024

23-200-204B BASIC ELECTRICAL ENGINEERING (2023 Scheme)

Branch – SFE, ME-A, ME-B, CE-A, CE-B

TIME: 2 hrs **MAX. MARKS: 30**

Course Outcomes

On completion of this course the student will be able to:

CO1	Explain the concepts of various energy sources and electric circuits.
CO2	Apply the basic electrical laws to solve circuits.
CO3	Discuss the construction and operation of various electrical machines.
CO4	Identify suitable electrical machine for practical implementation

PART A

(Answer all questions) **(5 X 2= 10 Marks)**

Q.No.	Questions	Marks	BL	CO	PO
I (a)	Define different types of powers	2	L1	2	1,2
(b)	The maximum flux density in the core of a 250/3000 V, 50 Hz single phase transformer is 1.2 Wb/m^2. If the emf per turn is 8V, determine the primary and secondary turns and the area of the core.	2	L3	4	1,2
(c)	Discuss the classification of self-excited dc motors	2	L1	3	1,2
(d)	A four-pole generator, having wave-wound armature winding has 51 slots, each slot containing 20 conductors. What will be the voltage generated in the machine when driven at 1500 rpm assuming the flux per pole to be 7.0 mWb ?	2	L2	3	1,2
(e)	Describe the importance of back emf in dc motor	2	L1	3	1,2

PART B

(Answer the following questions choosing either the first OR second option)

(2 X 10= 20 Marks)

Q.No.	Questions	Marks	BL	CO	PO
II (a)	Explain the operation of a single-phase transformer and derive its emf equation	5	L1	4	1,2
(b)	Voltage and current for a circuit with two elements in series are expressed as $v(t) = 170 \sin(6280t + \frac{\pi}{3})$	5	L3	4	1,2

	$i(t) = 8.5 \sin(6280t + \frac{\pi}{2})$ (i) Plot the two waveforms (ii) Determine the frequency in Hz (iii) Determine power factor (iv) What are the values of those elements connected.				
	OR				
III	The efficiency of a 400kVA, 1ø transformer is 98.7% when delivering full load at 0.8 power factor lagging and 99.1% at half load at unity power factor Calculate a) Iron loss b) Full load copper loss	10	L3	2	1,2
IV	With a neat diagram explain the construction and working of a dc generator.	10	L2	3	1,2
	OR				
V (a)	A six-pole lap wound dc generator has 51 slots, each slot containing 18 conductors. The flux per pole is 35 mWb. Find the generated emf in the armature, if it is driven at a speed of 750 rpm.	5	L2	3	1,2
(b)	A three-phase delta connected balanced load connected to 400 V supply draws 17.32 A at 0.8 lagging. Each phase has a resistor and an inductor connected in series. Find the resistance and reactance per phase.	5	L2	2	1,2

Bloom's Taxonomy Levels

L1 - %	L2- %	L3-%	L4-%	L5-%	L6-%
22	44	34	0	0	0